

Ex.No: EXPLORATORY DATA ANALYSIS (EDA) – DATA CLEANING AND DATA TRANSFORMATION

Date:

AIM:

ALGORITHM:

CODE:

```
import pandas as pd
```

```
df=pd.read_csv("C:/Users/DELL/Downloads/Sleep_health_and_lifestyle_dataset.csv")
```

```
print(df.isnull())
```

	Person ID	Gender	Age	Occupation	Sleep Duration	Quality of Sleep \
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
..	...	...	...	...	...	...
369	False	False	False	False	False	False
370	False	False	False	False	False	False
371	False	False	False	False	False	False
372	False	False	False	False	False	False
373	False	False	False	False	False	False

	Physical Activity	Level	Stress	Level	BMI Category	Blood Pressure \
0		False		False	False	False
1		False		False	False	False
2		False		False	False	False
3		False		False	False	False
4		False		False	False	False
..		...		...	...	...
369		False		False	False	False
370		False		False	False	False
371		False		False	False	False
372		False		False	False	False
373		False		False	False	False

	Heart Rate	Daily Steps	Sleep Disorder
0	False	False	True
1	False	False	True
2	False	False	True
3	False	False	False
4	False	False	False
..	...	...	...
369	False	False	False
370	False	False	False
371	False	False	False
372	False	False	False
373	False	False	False

```
[374 rows x 13 columns]
```

```
print(df.isnull().sum())
```

Person ID	0
Gender	0
Age	0
Occupation	0
Sleep Duration	0
Quality of Sleep	0
Physical Activity Level	0
Stress Level	0
BMI Category	0
Blood Pressure	0
Heart Rate	0
Daily Steps	0
Sleep Disorder	219
dtype: int64	

```
df["Age"].fillna(df["Age"].mean(), inplace=True)
```

```
print(df)
```

	Person ID	Gender	Age	Occupation	Sleep Duration \
0	1	Male	27	Software Engineer	6.1
1	2	Male	28	Doctor	6.2
2	3	Male	28	Doctor	6.2
3	4	Male	28	Sales Representative	5.9
4	5	Male	28	Sales Representative	5.9
...	...	...	...	...	...
369	370	Female	59	Nurse	8.1
370	371	Female	59	Nurse	8.0
371	372	Female	59	Nurse	8.1
372	373	Female	59	Nurse	8.1
373	374	Female	59	Nurse	8.1

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
0	6	42	6	Overweight
1	6	60	8	Normal
2	6	60	8	Normal
3	4	30	8	Obese
4	4	30	8	Obese
...	...	...	...	...
369	9	75	3	Overweight
370	9	75	3	Overweight
371	9	75	3	Overweight
372	9	75	3	Overweight
373	9	75	3	Overweight

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea
...	...	...	...	...
369	140/95	68	7000	Sleep Apnea
370	140/95	68	7000	Sleep Apnea
371	140/95	68	7000	Sleep Apnea
372	140/95	68	7000	Sleep Apnea
373	140/95	68	7000	Sleep Apnea

```
[374 rows x 13 columns]
```

```
print(df.duplicated())
```

```
df.drop_duplicates(inplace=True)
```

```
print(df)
```

```

0      False
1      False
2      False
3      False
4      False
...
369    False
370    False
371    False
372    False
373    False
Length: 374, dtype: bool

```

	Person ID	Gender	Age	Occupation	Sleep Duration \
0	1	Male	27	Software Engineer	6.1
1	2	Male	28	Doctor	6.2
2	3	Male	28	Doctor	6.2
3	4	Male	28	Sales Representative	5.9
4	5	Male	28	Sales Representative	5.9
...	...	...	...	...	...
369	370	Female	59	Nurse	8.1
370	371	Female	59	Nurse	8.0
371	372	Female	59	Nurse	8.1
372	373	Female	59	Nurse	8.1
373	374	Female	59	Nurse	8.1

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
0	6	42	6	Overweight
1	6	60	8	Normal
2	6	60	8	Normal
3	4	30	8	Obese
4	4	30	8	Obese
...	...	...	...	...
369	9	75	3	Overweight
370	9	75	3	Overweight
371	9	75	3	Overweight
372	9	75	3	Overweight
373	9	75	3	Overweight

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
1	125/80	75	10000	NaN
2	125/80	75	10000	NaN
3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea
...	...	...	...	...
369	140/95	68	7000	Sleep Apnea
370	140/95	68	7000	Sleep Apnea
371	140/95	68	7000	Sleep Apnea
372	140/95	68	7000	Sleep Apnea
373	140/95	68	7000	Sleep Apnea

[374 rows x 13 columns]

```
Q1 = df['Age'].quantile(0.25)
```

```
Q3 = df['Age'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
```

```
upper = Q3 + 1.5 * IQR
```

```
outliers = df[(df['Age'] < lower) | (df['Age'] > upper)]
```

```
print(outliers)
```

```
Empty DataFrame
Columns: [Person ID, Gender, Age, Occupation, Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, BMI Category, Blood Pressure, Heart Rate, Daily Steps, Sleep Disorder]
Index: []
```

```
Q1 = df['Sleep Duration'].quantile(0.25)
```

```
Q3 = df['Sleep Duration'].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
```

```
upper = Q3 + 1.5 * IQR
```

```
outliers = df[(df['Sleep Duration'] < lower) | (df['Sleep Duration'] > upper)]
```

```
print(outliers)
```

```
Empty DataFrame
Columns: [Person ID, Gender, Age, Occupation, Sleep Duration, Quality of Sleep, Physical Activity Level, Stress Level, BMI Category, Blood Pressure, Heart Rate, Daily Steps, Sleep Disorder]
Index: []
```

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
df[['Age']] = scaler.fit_transform(df[['Age']])
```

```
print(df[['Age']])
```

```
      Age
0  0.00000
1  0.03125
2  0.03125
3  0.03125
4  0.03125
..     ...
369 1.00000
370 1.00000
371 1.00000
372 1.00000
373 1.00000
```

```
[374 rows x 1 columns]
```

```
from sklearn.preprocessing import LabelEncoder
```

```
encoder = LabelEncoder()
```

```
df['Gender'] = encoder.fit_transform(df['Gender'])
```

```
print(df)
```

	Person ID	Gender	Age	Occupation	Sleep Duration \
0	1	1	0.00000	Software Engineer	6.1
1	2	1	0.03125	Doctor	6.2
2	3	1	0.03125	Doctor	6.2
3	4	1	0.03125	Sales Representative	5.9
4	5	1	0.03125	Sales Representative	5.9
..	...	...	...	...	...
369	370	0	1.00000	Nurse	8.1
370	371	0	1.00000	Nurse	8.0
371	372	0	1.00000	Nurse	8.1
372	373	0	1.00000	Nurse	8.1
373	374	0	1.00000	Nurse	8.1

	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category \
0	6	42	6	Overweight
1	6	60	8	Normal
2	6	60	8	Normal
3	4	30	8	Obese
4	4	30	8	Obese
..	...	...	...	...
369	9	75	3	Overweight
370	9	75	3	Overweight
371	9	75	3	Overweight
372	9	75	3	Overweight
373	9	75	3	Overweight

	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder
0	126/83	77	4200	NaN
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3	140/90	85	3000	Sleep Apnea
4	140/90	85	3000	Sleep Apnea
..	...	...	...	...
369	140/95	68	7000	Sleep Apnea
370	140/95	68	7000	Sleep Apnea
371	140/95	68	7000	Sleep Apnea
372	140/95	68	7000	Sleep Apnea
373	140/95	68	7000	Sleep Apnea

[374 rows x 13 columns]

pd.get_dummies(df['Gender'])

	0	1
0	False	True
1	False	True
2	False	True
3	False	True
4	False	True
...	...	...
369	True	False
370	True	False
371	True	False
372	True	False
373	True	False

374 rows × 2 columns

RESULT: